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Parkinson disease detection based on in-air dynamics feature extraction and selection using machine learning

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Parkinson's disease (PD) is a progressive neurological disorder that impairs movement control, leading to symptoms such as tremors, stiffness, and bradykinesia. Early and accurate PD detection is essential for effective management and improving patient outcomes. Many researchers analyzing handwriting data for PD detection typically rely on computing statistical features over the entirety of the handwriting task. While this method can capture broad patterns, it has several limitations, including a lack of focus on dynamic change, oversimplified feature representation, a lack of directional information, and missing micro-movements or subtle variations. Consequently, these systems face challenges in achieving good performance accuracy, robustness, and sensitivity. To overcome this problem, we proposed an optimized PD detection methodology that incorporates newly developed dynamic kinematic features and machine learning (ML)—based techniques to capture movement dynamics during handwriting tasks. Unlike typical Parkinson's Disease (PD) detection methods, which only differentiate between PD and non-PD cases, our approach classifies PD patients into distinct stages—early, mid, and late—based on the age of the disease, reflecting its progression over time. In the procedure, we first extracted 65 newly developed kinematic features from the handwriting task, aiming to bring significant variations in acceleration, deceleration, and directional changes—subtle movements that traditional methods may struggle to detect. We also reused 23 existing kinematic features, resulting in a comprehensive new feature set. Next, we enhanced the kinematic features by applying statistical formulas to compute hierarchical features from the handwriting data. This approach allows us to capture subtle movement variations that distinguish PD patients from healthy controls. To further optimize the feature set, we applied the Sequential Forward Floating Selection method to select the most relevant features, reducing dimensionality and computational complexity. Finally, we employed an ML-based approach based on ensemble voting across top-performing tasks, achieving an impressive 96.99% accuracy on task-wise classification and 99.98% accuracy on task ensembles, surpassing the existing state-of-the-art model by 2% for the PaHaW dataset. This exceptional accuracy underscores the transformative potential of our approach in redefining the benchmarks for PD detection. Our code and data are available at: https://github.com/musaru/PD_PaHaW.

Keywords Parkinson's disease, PaHaW dataset, Computer-aided disease recognition, Handwriting, Kinematic features, Dynamic movement, Decision support system, Features extraction, SFFS, Early stage, Mid stage, Late stage PD, Machine learning, Feature selection

Parkinson's disease (PD) is a prevalent neurodegenerative disorder, affecting approximately 6 million people worldwide in 2016, with prevalence rates expected to rise as the population ages[1–3]. While PD typically affects those over 60, juvenile PD can develop before age 40, often due to genetic factors. The disease results from a lack of dopamine, a chemical responsible for controlling movement. As PD progresses, it impacts various brain regions, with non-motor symptoms often appearing years before the disease is diagnosed. Currently, there is no cure for

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PD, and treatment options are aimed at managing symptoms and slowing disease progression. Early diagnosis is critical to prevent further deterioration. Clinically, PD severity is commonly measured using the Unified Parkinson's Disease Rating Scale (UPDRS), but the diagnosis is not always accurate. Studies have shown that around 25% of PD cases are misdiagnosed, even by movement disorder specialists[4,5]. In this scenario, automated PD detection is essential to improve patients' lives and enhance the accuracy of PD diagnostic systems. The primary symptoms of PD include tremors, muscle stiffness, slow movement (akinesia), and balance issues. In addition, sudden freezing of movements is also a crucial attribute to provide important diagnostic clues. Many researchers focus on video-based movement analysis or MRI analysis to develop PD[1,5,6] or movement disorder detection using machine learning and deep learning models[7–12]. However, due to high computational complexity with the large scale dataset researchers focus on handwriting analysis and this consider as promising method for detecting PD, as handwriting[13] changes are often an early indicator of the disease. However, due to the high computational complexity of large-scale video or MRI datasets, researchers focus on handwriting analysis, which is a useful method for early PD detection. Handwriting changes, such as smaller letters (micrographia)[14], tremors, slower movement[15], and longer writing times[16], serve as important kinematic features[14,17]. By analyzing these features, which vary with PD severity, researchers apply machine learning algorithms to achieve accurate and efficient classification for real-time automated PD diagnosis. Among them, Drotár et al.[18] developed a PD recognition system using 13 features, including trajectory, velocity, jerk, and stroke. By applying an SVM with an RBF kernel, they achieved an accuracy of 79.4% in distinguishing PD patients from healthy controls. Their later study[19] expanded to 600 features and improved classification accuracy to 80.09%. In another study, Drotár et al.[20] employed entropy, signal energy, and empirical mode decomposition (EMD) as features, achieving an accuracy of 88.1% In a similar study, Mucha et al.[21] included features such as velocity, acceleration, frequency domain features, and fractional-order derivatives, achieving an accuracy of 97.14% using XGBoost for classification. Impedovo et al.[22] used 24 baseline features and an SVM with linear kernel, achieving an accuracy of 88.33%, which improved to 93.79% with the addition of new features. When focusing on three tasks, they achieved an accuracy of 97.14%.

Despite these contributions, most existing methods rely on features extracted from the entire handwriting task, which overlooks the dynamic changes that occur during specific parts of the task[18–23]. The initial and final stages of handwriting often exhibit significant variations in acceleration, deceleration, and directionality—subtle movements that are challenging to capture using conventional methods. Moreover, there is a lack of focus on dynamic changes, oversimplified feature representation, insufficient directional information, and neglect of micro-movements or subtle variations. Consequently, existing systems struggle with performance accuracy, robustness, and sensitivity due to lacking effective features. To address these challenges, we proposed an optimized PD detection methodology that incorporates newly developed dynamic kinematic features and machine learning techniques to capture movement dynamics during handwriting tasks. We use the newly developed movement dynamic 65 kinematic feature to capture the subtle movement changes occurring during the initial and final stages of writing, besides the full signals, which may differ significantly between PD patients and healthy controls (HC). Our main contributions to this study are given below:

- New dynamic feature extraction method: We extracted 65 new kinematic features to extract the significant variations in acceleration, deceleration, and directional changes—subtle movements that conventional methods may struggle to detect. These features include angle trajectory, signed x/y displacement, velocity, first/last displacement, etc. Moreover, we introduce angle features related to handwriting and a directional speed feature, which accounts for the movement direction rather than relying solely on the absolute speed values in the x- and y-axes. Alongside these 65 newly introduced features, we reused 23 existing kinematic features, resulting in a comprehensive feature set designed to improve PD detection accuracy. As we are using the first and last phases of the handwriting tasks, it also makes completely new features. We extracted these features to address the limitations in existing feature sets for detecting PD-related movement characteristics across multiple tasks.
- Hierarchical feature extension: We enhanced the kinematic features by applying statistical theorems, including central tendency, dispersion, and higher-order relationships, to compute hierarchical features from the handwriting data. This hierarchical feature leads to gaining deeper insights into the writing process from each Kinematic feature. By integrating the newly extracted features with the existing ones, we calculated around $(65 + 23) \times (11 \pm 3) = 844$ comprehensive and practical feature set that significantly improves the detection of PD-specific movement patterns during handwriting tasks.
- Feature selection with sequential forward floating selection (SFFS): We employ SFFS to identify the most impactful robust features, refining the feature set to ensure optimal performance across different ML-based classifiers. This selection process enhances the system's ability to focus on the most relevant features for PD detection.
- Classifier optimization with optuna and ensemble voting: We employed leave-one-out-cross validation (LOOCV) for splitting the PaHaW dataset into training and testing. By leveraging Optuna for classifier optimization, we fine-tuned ML-based models for PD detection. Furthermore, ensemble voting across top-performing tasks increased the robustness and accuracy of our system, achieving an outstanding performance accuracy of 96.99% and 99.98% accuracy for individual task accuracy average and tasks ensemble, respectively, for the PaHaW dataset. Additionally, our method classifies PD patients into distinct stages—early, mid, and late—based on the age of the disease, enhancing the understanding of PD's evolution over time. This contrasts with typical PD detection methods, which only differentiate between PD and non-PD. This demonstrates the effectiveness and reliability of our approach, setting a new benchmark in PD detection methodologies.

The paper is organized as follows. After the introductory section, the description of the used database is given. Next, the methodology of extracting the features from handwriting signals is given, followed by a brief overview of the used classifier. Finally, the numerical results and conclusions are provided. Our contributions to the study are given below:

Related works

The PD/HC classification problem in handwriting analysis has attracted significant attention, with many studies exploring how handwriting features can reveal motor impairments associated with Parkinson's Disease (PD). However, most existing methods rely on statistical features computed over the entire handwriting task, which may fail to capture the subtle dynamic changes critical for accurate PD detection. Drotár et al.[18] investigated various kinematic features, including stroke speed, velocity, acceleration, and jerk. They found strong correlations between certain feature vectors (e.g., stroke speed and stroke width) and PD responses. However, their study analyzed the overall handwriting task, which overlooked finer dynamics, such as initial movement patterns and final execution, both crucial for detecting early PD symptoms. To address this, Drotár et al.[19] included both on-surface and in-air trajectory data to improve PD prediction. They found that eight of the top ten most discriminative features were related to in-air trajectories, indicating that dynamic handwriting changes, including movements above the writing surface, could provide valuable insights into PD. While this expanded the feature set, the study still relied on analyzing the full handwriting task, limiting sensitivity to subtle changes in the initial and final stages. Moetesum et al.[24] introduced Convolutional Neural Networks (CNNs) to extract visual features from handwriting and used SVM classifiers to classify PD. While their method achieved 83% classification accuracy, it did not focus on dynamic changes within specific portions of the handwriting task, missing the opportunity to capture critical early or late-stage variations in PD patients' handwriting. Similarly,[20] incorporated entropy, energy, and empirical mode decomposition as additional features, achieving high classification accuracy using Relief-selected features and RBF kernel SVM. However, their method still relied on features extracted from the entire handwriting task, which may miss subtle dynamic changes in handwriting's initial and final phases. Mucha et al.[21] proposed the use of fractional-order derivative (FD) features for analyzing handwriting dynamics. Their study showed that FD-based features correlated more significantly with clinical characteristics (UPDRS V and PD duration) than traditional kinematic features. Using XGBoost for classification, they achieved an accuracy of 97.74%. However, this method also uses features extracted from the entire task, limiting its ability to capture nuances during specific stages of the handwriting process. Impedovo et al.[22] explored sigma-lognormal modeling, Maxwell-Boltzmann distribution, and discrete Fourier transforms for PD classification, achieving a classification accuracy of 98.44% by combining the top three best-performing tasks. While successful, their approach still relied on overall handwriting task features. Parziale et al.[25] introduced the Negative Selection Algorithm (NSA), which only required HC data for training, alleviating the burden of collecting patient data. Their method reported an average accuracy of 97.12%, but like other studies, it relied on features from the entire handwriting task. Krut et al.[26] focused on the Archimedean spiral task, using Dynamic Time Warping (DTW) to optimize matching between individual data and a reference spiral. Their SVM model achieved 97.52% accuracy, but again, their method did not isolate dynamic changes within the writing process, which could be crucial for PD detection. Deharab et al.[27] adopted Dynamic Writing Traces Warping (DWTW) combined with an SVM model, reporting a classification accuracy of 88.33% for Task 8 of the PaHaW dataset. However, like other approaches, it failed to capture the subtle, dynamic variations in handwriting during the first and last portions of the task.

Despite the valuable contributions of the studies mentioned above, most existing methods rely heavily on statistical features computed over the entire handwriting task[18–23]. While this approach captures general handwriting patterns, it has several limitations. The main problem of the existing feature extraction study mentioned above is that it often relies on statistical features computed over the entirety of the handwriting task. This entire handwriting task used for the feature extraction approach may capture broad patterns. However, it has significant limitations, including a lack of focus on dynamic changes, oversimplified feature representation, insufficient directional information, and neglect of micro-movements or subtle variations. Consequently, existing systems struggle with performance accuracy, robustness, and sensitivity. To address these limitations, we propose an optimized methodology for PD detection that focuses on the first 10% and last 10% of the handwriting task, alongside features such as kurtosis and skewness indices. These portions of the task capture dynamic handwriting changes that are sensitive to motor impairments in PD patients, which might otherwise be missed in full-task analysis. We also leverage newly developed movement dynamic features to capture subtle movement variations that differentiate PD patients from HC. By focusing on the first and last portions of handwriting and applying an ensemble learning approach with SVM classifiers for each feature, our methodology achieves 96.99% classification accuracy for task-wise evaluations and 99.98% accuracy for ensemble tasks, outperforming existing models by 2%.

PaHaW dataset

Experimental evaluation has been performed on the PaHaW dataset[28]. It includes 37 PD patients and 38 HC. Tasks include words written in Czech (the participants' native language). Each participant completed 8 handwriting tasks: 1. drawing the Archimedes spiral; 2. writing in cursive the letter "l", 3. the bigram "le" and 4. the trigram "les"; 5. writing in cursive the words "lektorka" (female teacher in Czech), 6. "porovnat" (to compare), and 7. "nepopadnout" (to not catch); 8. writing in cursive the sentence "Tramvaj dnes uz nepojede" (The tram won't go today). Table 1 shows the demographic information of the PaHaW dataset. The demographic data in the PaHaW dataset shows that PD patients have a mean disease duration of 8.37 years, with participants from various stages of the disease. The PaHaW dataset includes 75 participants, of which 38 are healthy controls and

Parameter	PD patients (n = 37)	Healthy controls (n = 38)
Age (years)	Mean: 69.3, SD: 10.9	Mean: 62.4, SD: 11.3
Sex	19 male / 18 female	20 male / 18 female
Dominant hand	Right handed: 37	Right handed: 38
Years of education	Mean: 10+ years	Mean: 10+ years
Disease duration (years)	Mean: 8.37, SD: 4.8	N/A
Age of onset (years)	Mean: 60.0, SD: 8.5	N/A
Hoehn , Yahr stage	Stage 1: 12, stage 2: 15, stage 3: 10	N/A
UPDRS part V score	Mean: 2.27, SD: 0.84	N/A
Levodopa equivalent daily dose (mg)	Mean: 1373.4, SD: 714	N/A
Cognitive scores (MMSE)	Mean: 28.5, SD: 1.2	Mean: 29.2, SD: 0.8
Cognitive scores (MoCA)	Mean: 24.1, SD: 3.0	Mean: 25.3, SD: 2.5

Table 1. Demographic and clinical data of participants[28].

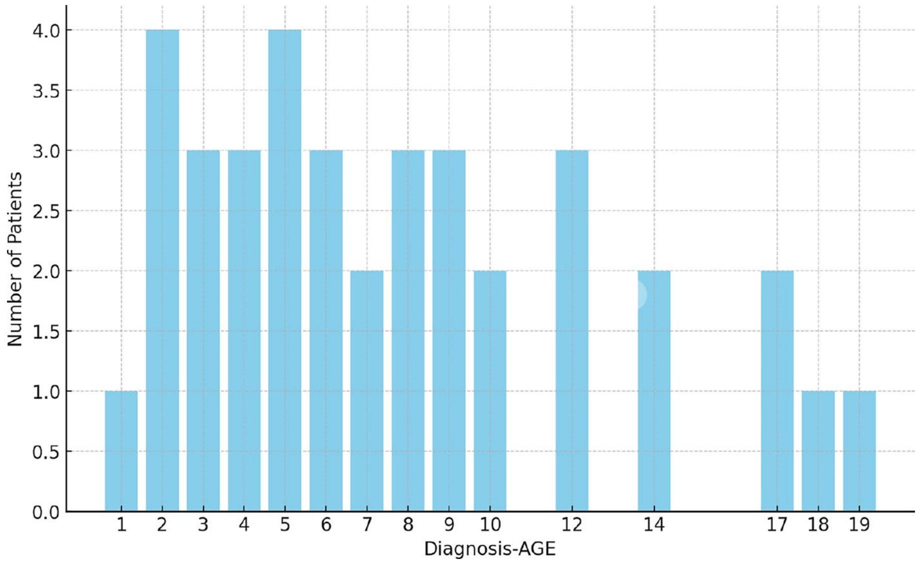

Fig. 1. Diagnosis age distribution of the PaHaW dataset[28].

37 are PD patients. The PD patients have an average disease duration of 8.37 years, with participants spanning various stages of Parkinson's Disease (PD). The breakdown of patients by diagnosis age is as follows shown in Fig. 1: 11 patients with 1 to 4 years of diagnosis (likely in Stage 1, representing early PD), 15 patients with 5 to 9 years of diagnosis (likely in Stage 2, representing moderate PD), and the remaining 11 patients with 10+ years of diagnosis, corresponding to Stage 3 (mid-to-late PD). Therefore, the dataset includes 11 patients in Stage 1, 15 patients in Stage 2, and 11 patients in Stage 3. The group in Stage 1 (early PD) along with those in Stage 2 were included in the evaluation of our detection methods for early and moderate stages of PD. However, the majority of participants are in the mid-to-later stages (Stages 2 and 3), and therefore, conclusions regarding the effectiveness of the methods for detecting early PD should be interpreted with this in mind.

The data collection procedure included a digitising tablet (Wacom Intuos 4M) that was overlaid with an empty paper template, and participants were allowed to repeat a task if they made mistakes. Online handwriting signals were recorded with the 150 Hz sampling rate fs. The raw data captured by the device are the x- and y-y-coordinates of the pen position and their timestamps. Moreover, measures of pen inclination, i.e., azimuth and altitude, and the pressure were recorded. The last signal concerns the so-called button status, which is a binary variable evaluating 0 for the pen-up state (in-air movement) and 1 for the pen-down state (on-surface movement). In this context, this dataset collection method adopted a Pen-Tablet device for analyzing PD. The main advantage of online acquisition devices is their ability to acquire the kinematics (dynamics) of the writing process, which are lost in offline systems. In this case, the trait is represented as a sequence $S(n)_{n=0,1,\ldots,N}$, where $S(n)$ is the signal value sampled at time $n\Delta t$ of the writing process ($0 \leq n \leq N$), Δt is the sampling period.

Materials and methods

The overall system workflow architecture is illustrated in Fig. 2. Despite significant efforts by researchers to improve performance accuracy using the benchmark PaHaW handwriting dataset, challenges remain in

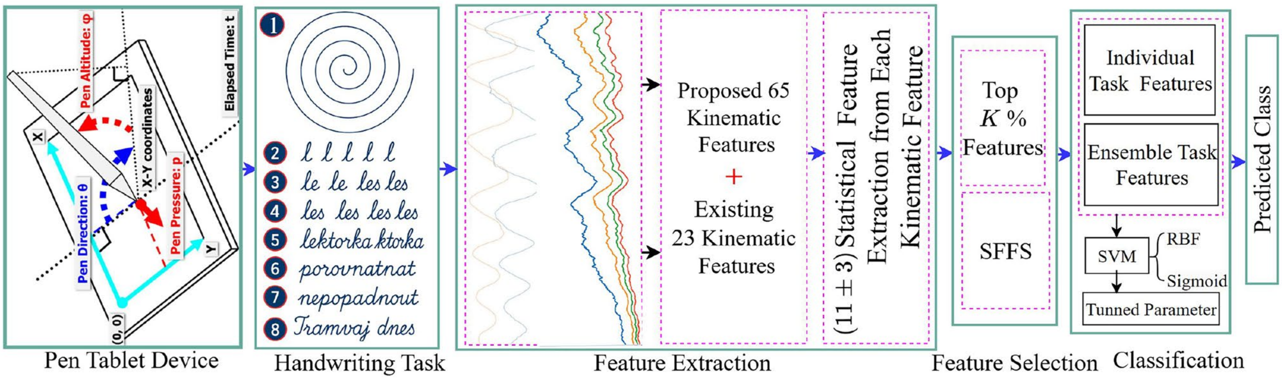

Fig. 2. Our proposed model architecture for PD detection.

achieving optimal results[18–23]. These issues arise from a lack of focus on dynamic changes, oversimplified feature representation, insufficient directional information, and neglect of micro-movements or subtle variations. As a result, existing systems struggle with performance accuracy, robustness, and sensitivity. To address these challenges, we propose an optimized methodology for PD detection, incorporating newly developed dynamic kinematic features and machine learning techniques to capture the movement dynamics during handwriting tasks. The system consists of three main stages:

- Stage 1: feature extraction
  In this stage, we newly proposed 65 dynamic kinematic features extraction approaches where some features were extracted from the entire handwriting task and some features were extracted from the first and last 10% phases of the handwriting task instead of using the whole task. These phases are crucial because they exhibit significant variations in acceleration, deceleration, and directional changes—subtle movements that conventional methods often fail to detect. By isolating these segments, our approach captures critical transitions and movement irregularities that are more pronounced in PD patients. The extracted features include angle trajectory, signed x/y displacement, velocity, first/last displacement, and others. We also introduce new angle and directional speed features, which capture movement direction rather than relying solely on absolute speed values in the x- and y-axes. Additionally, we incorporated 23 existing kinematic feature extraction methods, enhanced by focusing on the initial and final phases of the handwriting task, providing a completely new set of features.
- Stage 2: hierarchical feature extension
  We extended the kinematic features by applying statistical theorems, including central tendency, dispersion, and higher-order relationships, to compute hierarchical features from the handwriting data. This enables deeper insights into the writing process, enhancing our understanding of PD-specific movement patterns. The resulting comprehensive feature set, consisting of both newly developed and existing features, is augmented with approximately 844 statistical features derived from the original 88 kinematic features.
  Stage 3: feature selection and model optimization
  To reduce computational complexity and improve performance, we applied Sequential Forward Floating Selection (SFFS) to identify the most informative features, minimizing redundancy and ensuring optimal performance across different classifiers. Additionally, SVM hyperparameters were optimized using Optuna to enhance classifier robustness. Finally, ensemble voting was performed across top-performing tasks, including top-3, top-5, and all tasks, to evaluate and maximize system performance.

This approach ensures the system achieves high accuracy, sensitivity, and robustness in detecting PD, outperforming existing state-of-the-art methods. The details description of the each procedure are given below.

Feature extraction

In this study, we focus on enhancing the accuracy and robustness of PD detection by extracting kinematic features and then extending them with statistical formulas to calculate hierarchical features from handwriting tabular data. While existing methods have used traditional kinematic features, such as pressure, azimuth, altitude, displacement, velocity, and stroke count, these often fail to capture subtle and nuanced movement patterns. Traditional kinematic features typically consider the tabular device information and their entire range of data, calculating values such as x, y, z, azimuth, altitude, displacement or velocity over the full task duration. However, such methods overlook the fine details of handwriting motion that can vary across different phases of the writing task. This limitation may reduce sensitivity when distinguishing between PD patients and healthy controls, especially in terms of micro-movements or variations in behaviour during specific phases of the handwriting task. To address these challenges, we introduce 65 new dynamic movement-based kinematic features in addition to the 23 baseline features from previous studies. These new features focus on capturing detailed variations in handwriting dynamics, which are critical for improving detection accuracy. Our proposed kinematic feature extraction approach specifically targets the mentioned limitations by focusing on key segments of the handwriting process. We introduce features that are computed from designated portions of the recorded data, such as the first 10% and last 10% of the writing sequence. These phases are particularly important as they often exhibit significant variations in acceleration, deceleration, and directional changes—subtle movements that are difficult to detect using conventional methods. By isolating these segments, our approach is able to capture critical transitions and irregularities in movement that are more pronounced in PD patients.

For example, angle trajectory, signed displacement and velocity variations track directional changes and speed fluctuations during writing, offering insights into spatial inconsistencies and motor control deficits that may differ between PD patients and healthy individuals. Additionally, first/last displacement highlights differences in starting and ending positions, which can indicate unique motor impairments seen in PD patients. These new features delve deeper into the dynamic aspects of handwriting, moving beyond general movement patterns and providing greater sensitivity to subtle variations. Table 2 summarizes the newly introduced features. By incorporating these into our analysis pipeline, we aim to improve not only classification accuracy but also model interpretability and robustness. These advanced kinematic features are designed to enhance the ability to detect subtle irregularities in handwriting, leading to better performance in PD detection systems. Table 3 provides an overview of the features adopted by state-of-the-art approaches, with more details on their implementation found in[4,14,18–20,22]. Here, we first extracted kinematic features, then we extended function-based features by evaluating statistical parameters such as mean, median, standard deviation, 1st and 99th percentiles, kurtosis, and skewness. This extensive level of the feature extraction approach leads to the robust representation of handwriting dynamics and improves the overall detection model.

Figure 3 illustrates the dynamic changes in handwriting patterns for (a) Task 1 and (b) Task 2, comparing PD patients with healthy controls. Also, Fig. 4 shows the dynamic change of (a) Task 7 and (b) Task 8. Each subplot presents time series plots of kinematic features such as acceleration, deceleration, or directional changes recorded during the handwriting task. For each task, separate plots are shown for a healthy control and a PD patient. These visualizations highlight subtle dynamic differences in handwriting behavior that may not be easily detected without proper feature extraction, thereby aiding in the differentiation between PD patients and healthy individuals. Figures 5, 6, and 7 show the boxplot of some newly proposed kinematic features with the Std statistical feature.

Our proposed kinematic feature extraction approaches

Table 2 summarizes the new kinematic features proposed in this paper, which are designed to capture subtle changes in movement, particularly during the initial and final stages of handwriting. Here, we extracted a Total of 65 new features, where 37 features were from 100% of the handwriting task, 13 features were from the first 10% and 15

No.	Feature name	Description
1	Angle in trajectory	1–10% mean, 10–99% percentile, 100% median and standard deviation
2–3	Signed x and y displacement	Signed displacement captures the directionality of the pen's movement. Formula: $d_i = \frac{x_{i+1}(y_{i+1}) - x_i(y_i)}{t_{i+1} - t_i},\ 1 \le i \le n-1$
4–5	Signed x and y velocity	Signed velocity in x/y directions
6–8	First pen information	First pressure, first Azimuth, first altitude for the first 10% of the handwriting task
9	First displacement	First displacement calculated using: $d_i = \frac{\sqrt{(x_{i+1}-x_i)^2 + (y_{i+1}-y_i)^2}}{t_{i+1}-t_i},\ 1 \le i \le n \times 0.1$
10	First velocity	$v_i = \frac{d_i}{t_{i+1}-t_i},\ 1 \le i \le n \times 0.1$
11–12	First x and y displacement	Magnitude of displacement in x/y direction for the first 10% of the handwriting task
13–14	First x and y velocity	First velocity in the x/y direction for the first 10% of the handwriting task
15–16	First signed x and y displacement	Signed displacement in x/y direction during the first 10% of the handwriting task
17–18	First signed x and y velocity	Signed velocity in x/y direction for the first 10% of the handwriting task
19	Last displacement	Last displacement calculated using: $d_i = \frac{\sqrt{(x_{i+1}-x_i)^2 + (y_{i+1}-y_i)^2}}{t_{i+1}-t_i},\ n - n \times 0.1 \le i \le n-1$
20	Last velocity	$v_i = \frac{d_i}{t_{i+1}-t_i},\ n - n \times 0.1 \le i \le n-1$
21–22	Last x and y displacement	Magnitude of displacement in x/y direction for the last 10% of the handwriting task
23–24	Last x and y velocity	Last velocity in x/y direction for the last 10% of the task
25–26	Last signed x and y displacement	Signed displacement in x/y direction for the last 10% of the handwriting task
27–28	Last signed x and y velocity	Signed velocity in x/y direction for the last 10% of the handwriting task
29–30	Last x and y	Last displacement in x/y direction for the last 10% of the handwriting task
31–33	Last pen information	Last pressure, last Azimuth, last altitude for the last 10% of the handwriting task
34–44	Stroke pressure	Stroke pressure statistics for each stroke (max, min, mean, median, variance, std, skewness, kurtosis)
44–54	Stroke pressure displacement	Stroke displacement statistics for each stroke (max, min, mean, median, variance, std, skewness, kurtosis)
55–65	Stroke pressure velocity	Stroke velocity statistics for each stroke (max, min, mean, median, variance, std, skewness, kurtosis)

Table 2. Proposed dynamic movement 65 features extraction method.

No.	Feature name	Description
1–2	Position	Position in terms of s(x,y) or Coordinates
3–	Button status	Movement in the air: b(t)=0 ,Movement on the pad: b(t)=1
4	Pressure	Pressure of the pen on the pad
5	Azimuth	Angle between the pen and the vertical plane on the pad
6	Altitude	Angle between the pen and the pad plane
7	Displacement	$d_i = \begin{cases} \dfrac{\sqrt[2]{(x_{i+1} - x_i)^2 + (y_{i+1} + y_i)^2}}{t_{i+1} - t_i} & 1 \leqq i \leqq n-1 \\ d_n - d_{n-1} & i = n \end{cases}$
8	Velocity	$v_i = \begin{cases} \dfrac{d_i}{t_{i+1} - t_i} & 1 \leqq i \leqq n-1 \\ v_n - v_{n-1} & i = n \end{cases}$
9–10	x and y displacement	Displacement in the x and y direction. The specific formula is as follows. $d_i = \begin{cases} \left\lvert \dfrac{x_{i+1}(y_{i+1}) - x_i(y_i)}{t_{i+1} - t_i} \right\rvert & 1 \leqq i \leqq n-1 \\ d_n - d_{n-1} & i = n \end{cases}$
11–12	x and y velocity	Velocity in the x and y direction
13	NCV	Number of changes of Velocity
14	NCA	Number of changes of Acceleration
15	NCP	Number of changes of Pressure
16	Stroke number	Number of stroke
17	Durations	in-air time, on-surface time, total task time, Ratio of time spent in-air/on-surface
18–19	Entropy	Shannon and Rényi operators applied on (x,y)
20–21	Energy	Conventional Energy and Teager-Kaiser energy
22	SNR	Signal-to-noise ratio of the horizontal/vertical component of the pen position
23	EMD	Empirical Mode Decomposition

Table 3. Existing baseline features extraction method.

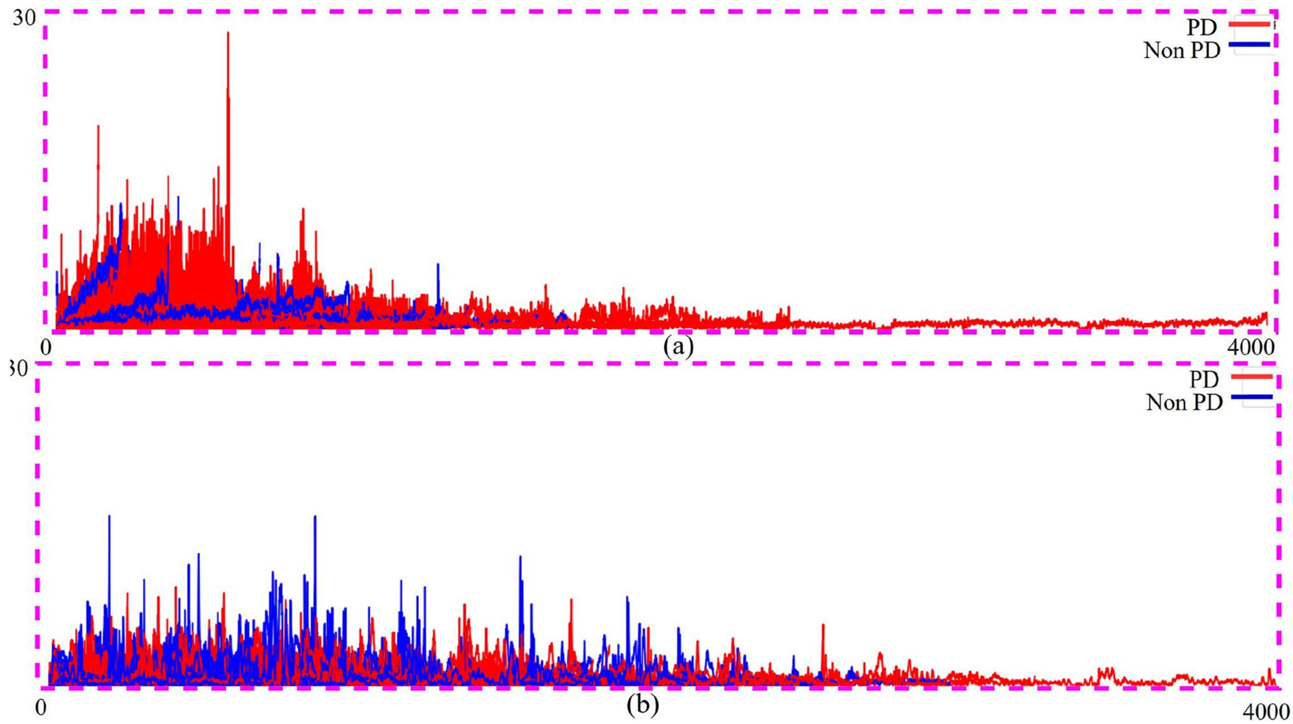

Fig. 3. Movement dynamics (time series plots): (**a**) Task 1 and (**b**) Task 2 visualize the dynamic changes in handwriting tasks for PD patients and healthy controls.

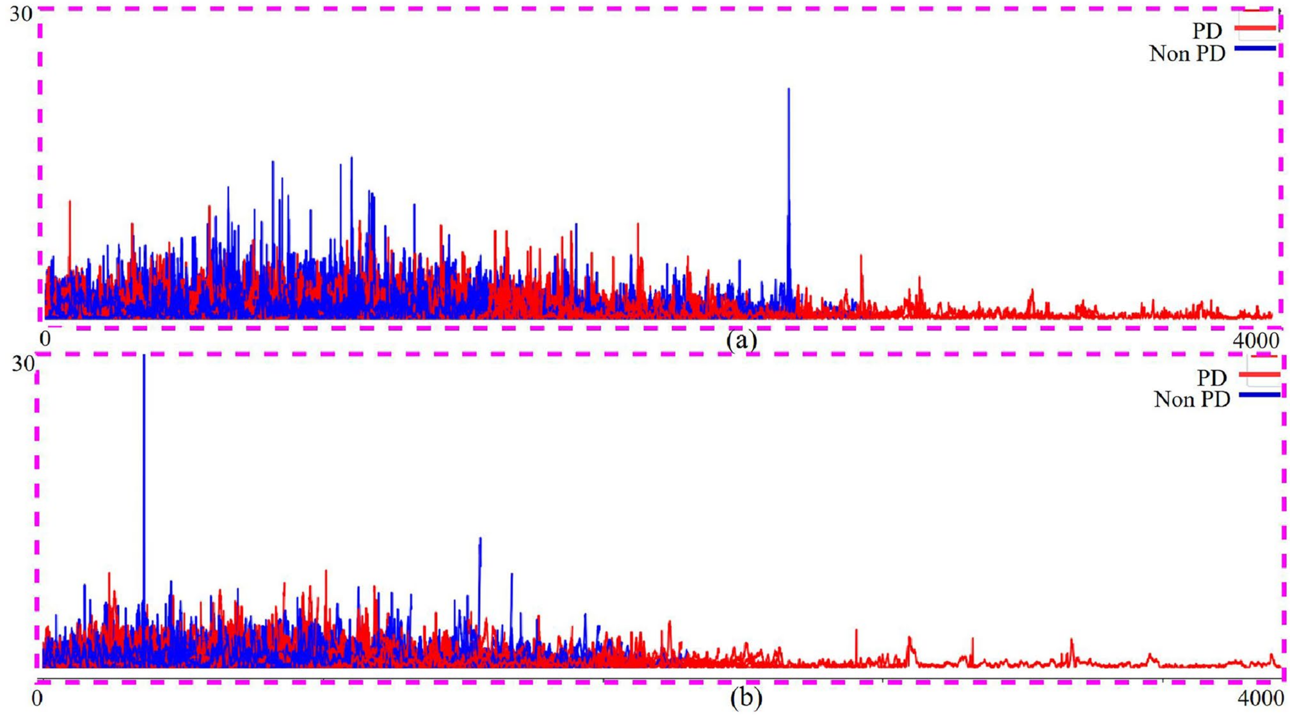

Fig. 4. Movement dynamics (time series plots): (**a**) Task 7 and (**b**) Task 8 visualize the dynamic changes in handwriting tasks for PD patients and healthy controls.

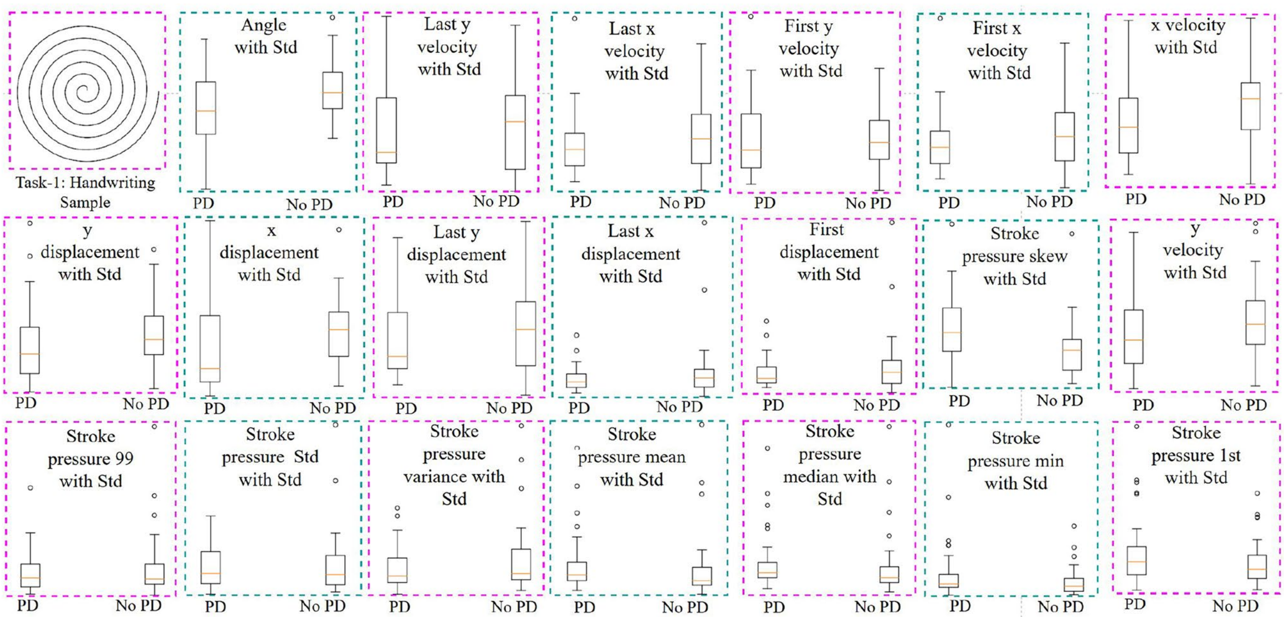

Fig. 5. Boxplot visualization of the proposed kinematic feature with Std statistical feature for Task-1.

features were from the last 10% of the handwriting task. Table 2 shows that features—1,2,3,4,5,34–44,44–54,55–65 were extracted from 100% of the handwriting task. Features No 6–8,9,10,11–18 extracted from the first 10% of the handwriting task. Feature No 19–33 extracted from the last 10% of the handwriting task. The decision to focus on the first and last 10% of the handwriting task was motivated by the idea that these portions of the task capture significant dynamic changes in handwriting that are particularly sensitive to motor impairments in PD patients. Specifically, the first 10% typically reflects the initiation of movement, which can be affected by bradykinesia (slowness of movement) and difficulty in initiating voluntary motor actions, both common early symptoms of PD. Similarly, the last 10% of the task captures the final execution of the handwriting, which

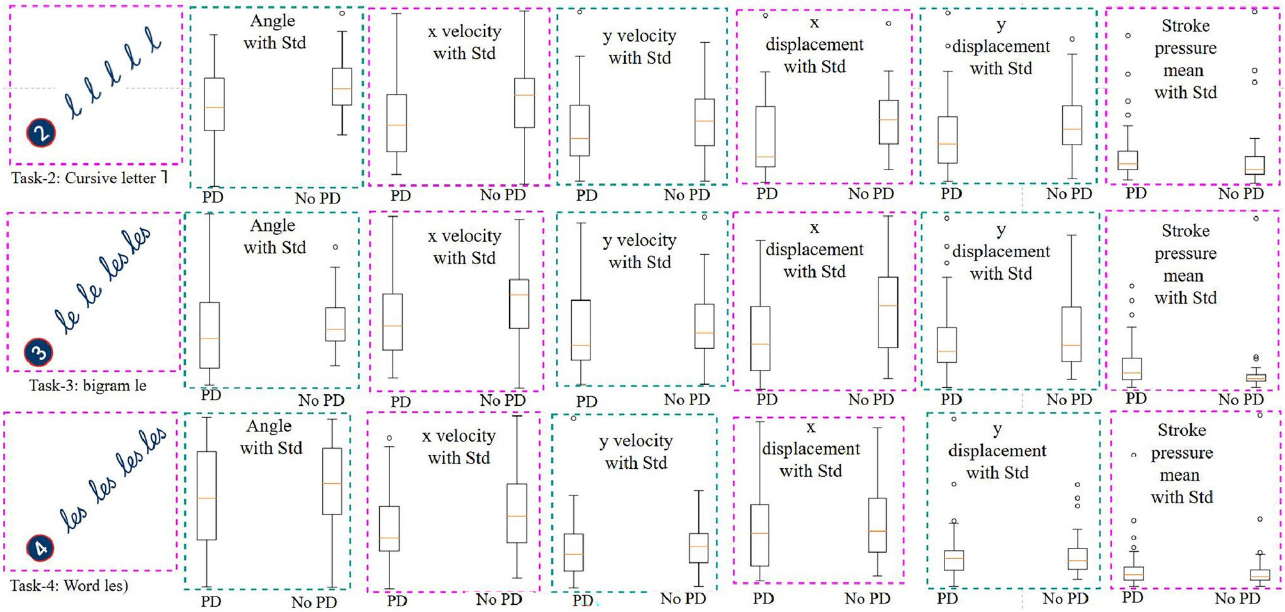

Fig. 6. Boxplot visualization of the proposed kinematic feature with Std statistical feature for Task-2,3,4.

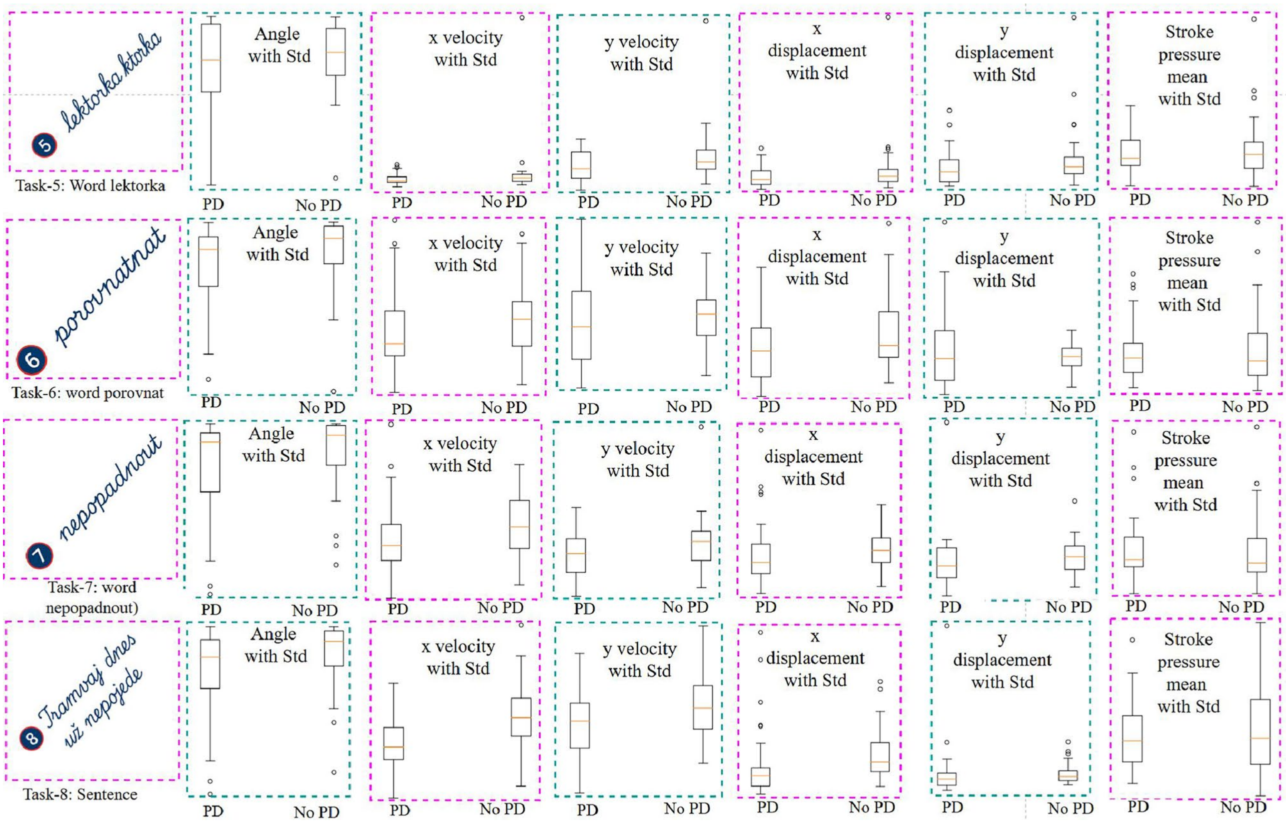

Fig. 7. Boxplot visualization of the proposed kinematic feature with Std statistical feature for Task-5–8.

can provide insight into fatigue, tremors, and other motor symptoms that may impact the end of the writing process. By focusing on these dynamic changes during handwriting, we aim to capture subtle variations in speed, pressure, and directionality that are not as easily detected when analyzing the entire task. In contrast, the full handwriting task (100%) captures the overall writing process and may average out these dynamic effects, making it harder to detect early signs of PD or motor variability in its initial and final phases. Therefore, isolating these critical portions of the handwriting task helps improve sensitivity and enhances the model's ability to detect PD-related abnormalities. For each of the first and last 10% of the task, we calculated features based on the writing movement in those specific portions (such as velocity, displacement, and pressure) and compared them to the remaining part of the handwriting task, which is analyzed using the full 100% of the handwriting data. The 100% corresponds to the entire writing task from start to finish, and it allows us to assess overall performance, including trends that may not be as evident in the segmented 10% portions. These features are enhanced with statistical parameters, enabling a deeper understanding of how handwriting dynamics differ between PD patients and HC. Specifically, these features provide valuable insights into how PD patients start and finish their handwriting tasks, revealing distinctive patterns that may be indicative of the disease. The newly introduced features are applied to key speed-related metrics, such as displacement and velocity, to track movement variations. By focusing on these dynamic aspects of handwriting, we aim to capture fine-grained differences that can help distinguish PD patients from healthy individuals. The new kinematic feature calculation procedure is described in detail below.

Angle in trajectory: first, we extracted the angle trajectory feature aiming to extract tremors and rigid-based information. Tremors and rigid muscles are well-known symptoms of PD. Even if the content is smooth for HC, such as straight lines and curves, PD may not be able to write smoothly due to these symptoms, and noise may be included in the trajectory. We thought we could identify it by calculating the curvature of the pen's trajectory to obtain this information.

Intuitively, we can decide any time *t*, and before time $t-1$, and after time $t+1$.The pen position at each time can be expressed as $p_t=(x_t,y_t)$, $p_{t-1}=(x_{t-1},y_{t-1})$ and $p_{t+1}=(x_{t+1},y_{t+1})$. Then we calculate two vectors ($\vec{v_1}=p_{t-1}-p_t$, and $\vec{v_2}=p_{t+1}-p_t$) with reference to time *t*. By using these two vectors and the inner product formula, the degree of curvature (angle) at time t can be calculated. Calculate angle formula is as follows:

$$\cos\theta=\frac{\vec{v_1}\cdot\vec{v_2}}{\|\vec{v_1}\|\|\vec{v_2}\|}$$

However, angle calculations at adjacent time intervals such as *t* and $t+1$ are often susceptible to small noises. So, by calculating the angle between the point of based time *t* and the point left any length(along the trajectory) from the point of based time *t*, we can change the magnitude of the noise affected. In this experiment, we calculated the angle from the based time *t* (based point p_t) to any point $d(px/nm/mm)\in(10,20,\ldots\ldots,100)$ in trajectory distance away.

Signed x / y displacement: this feature captures the signed displacement in the x and y directions, where the sign (+/-) retains information about the direction of movement. This distinction helps in understanding the directional aspects of the pen's movement during handwriting. The displacement formula is given by Eq. (1):

$$d_i=\frac{x_{i+1}(y_{i+1})-x_i(y_i)}{t_{i+1}-t_i},\quad 1\le i\le n-1 \tag{1}$$

Signed x/y velocity: this feature calculates the signed velocity in the x and y directions, accounting for the direction of movement. It helps in analyzing the speed dynamics in both horizontal and vertical axes, offering insights into handwriting anomalies.

First displacement: The first displacement represents the initial movement of the pen, capturing how far the pen moves in the early stages of the task. The formula is given by Eq. 2:

$$d_i=\frac{\sqrt{(x_{i+1}-x_i)^2+(y_{i+1}+y_i)^2}}{t_{i+1}-t_i},\quad 1\le i\le n\times 0.1 \tag{2}$$

First velocity: first velocity focuses on the speed at the beginning of the handwriting task, reflecting how quickly the pen moves during the initial movements. The velocity is computed using Eq. 3:

$$v_i=\frac{d_i}{t_{i+1}-t_i},\quad 1\le i\le n\times 0.1 \tag{3}$$

First x / y displacement: this feature calculates the magnitude of displacement specifically in the x and y directions, only for the first 10% of the total movement. See Eq. (4):

$$d_i=\left|\frac{x_{i+1}(y_{i+1})-x_i(y_i)}{t_{i+1}-t_i}\right|,\quad 1\le i\le n\times 0.1 \tag{4}$$

First signed x / y displacement: similar to the first displacement, but with the signed value retained, offering directionality information for the initial movements in the x and y directions. The formula is given by Eq. (5):

$$d_i=\frac{x_{i+1}(y_{i+1})-x_i(y_i)}{t_{i+1}-t_i},\quad 1\le i\le n\times 0.1 \tag{5}$$

Last displacement: this feature measures the displacement during the final stage of the handwriting task, analyzing how far the pen moves toward the end. The formula is provided in Eq. 6:

$$d_i = \frac{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} + y_i)^2}}{t_{i+1} - t_i}, \quad n - n \times 0.1 \le i \le n - 1 \tag{6}$$

Last velocity: this measures the velocity during the last 10% of the handwriting task, providing insight into the speed with which the writing is completed. The formula is given by Eq. 7:

$$v_i = \frac{d_i}{t_{i+1} - t_i}, \quad n - n \times 0.1 \le i \le n - 1 \tag{7}$$

Last x/y displacement: this feature computes the magnitude of displacement in the x and y directions for the final part of the task, as described by Eq. 8:

$$d_i = \left| \frac{x_{i+1}(y_{i+1}) - x_i(y_i)}{t_{i+1} - t_i} \right|, \quad n - n \times 0.1 \le i \le n - 1 \tag{8}$$

Last x/y velocity: this feature calculates the velocity in the x and y directions during the last 10% of the task, providing insights into the dynamics of pen movement at the final stage of handwriting. This velocity is given by Eq. (9):

$$v_i = \frac{d_i}{t_{i+1} - t_i}, n - n \times 0.1 \le i \le n - 1 \tag{9}$$

Last signed x/y displacement: this captures the signed displacement in the x and y directions at the end of the handwriting task, indicating how the pen moves directionally in its final stages. The formula is expressed by Eq. (10):

$$d_i = \frac{x_{i+1}(y_{i+1}) - x_i(y_i)}{t_{i+1} - t_i}, \quad n - n \times 0.1 \le i \le n - 1 \tag{10}$$

Last signed x/y velocity: this feature computes the signed velocity in the x and y directions during the final portion of the task, emphasizing the direction and speed of pen movement. The signed velocity formula is expressed by Eq. (11):

$$v_i = \frac{d_i}{t_{i+1} - t_i}, n - n \times 0.1 \le i \le n - 1 \tag{11}$$

Last pen information: this group of features captures the final pen states, including the pressure, azimuth, and altitude at the end of the handwriting task. These values can provide further insights into how the pen's physical parameters behave as the writing task concludes.

Stroke pressure: this group of features focuses on the pressure applied during each stroke. It includes several statistical parameters such as maximum, minimum, mean, median, variance, standard deviation, 1st and 99th percentiles, skewness, and kurtosis. These features provide a comprehensive view of the pressure dynamics during the handwriting task.

Stroke pressure displacement: this set of features focuses on the displacement of the pen during each stroke, incorporating statistical metrics such as maximum, minimum, mean, median, variance, standard deviation, 1st and 99th percentiles, skewness, and kurtosis. These features provide insights into how displacement patterns vary across individual strokes.

Stroke pressure velocity: this feature focuses on the velocity of the pen during each stroke, including statistical measures such as maximum, minimum, mean, median, variance, standard deviation, 1st and 99th percentiles, skewness, and kurtosis. It is designed to capture the speed variations during the strokes, offering further detail on how velocity fluctuates across different sections of the handwriting task.

Existing kinematic feature extraction approaches

In this study, we also employed 23 existing kinematic feature extraction methods, as shown in Table 3. These methods were implemented to serve as the baseline system for our analysis.

According to the Table 3 the 23 kinematic features used in this study encompass a range of metrics that capture both the spatial and temporal aspects of handwriting movement. These features provide a comprehensive view of the pen's behavior during writing tasks, including its position, displacement, and velocity in both horizontal (x) and vertical (y) directions. Features such as pressure, azimuth, and altitude give additional insights into the pen's interaction with the writing surface, reflecting how PD patients may exhibit altered pen grip, pressure variability, and orientation during handwriting. In addition, the number of changes in velocity, acceleration, and pressure track the irregularities in handwriting motion, which may be indicative of tremors, motor control deficiencies, or inconsistencies in movement.

Temporal aspects of the writing process, such as stroke number, durations, and the ratio of in-air time to on-surface time, further enhance the analysis by providing insights into the pacing and fluidity of handwriting. These features, when combined with measures like energy, entropy, and signal-to-noise ratio, allow for a more nuanced understanding of the motor impairments that can occur in PD. In particular, features related to entropy and SNR help quantify the complexity of movement and the influence of noise, while energy captures the intensity of pen motion, which may decrease in PD patients due to reduced motor vigour. These kinematic features, together with their corresponding statistical measures, serve as a powerful set of tools for distinguishing between normal and abnormal handwriting patterns in PD, providing a clearer picture of the subtle motor dysfunctions that may characterize the disease. These baseline features provide a general understanding of handwriting patterns but may lack sensitivity to more nuanced variations, particularly those that can distinguish between PD patients and HC. To over come the gap we proposed new 65 kinematic features.

Statistical feature extraction from each kinematic feature

In this study, we extracted 88 kinematic features, of which 65 are newly proposed and 23 are from existing studies. To capture subtle details of each task and movement pattern, we derived 11 ± 3 statistical features from each of the kinematic features. These statistical features can be categorized into two types: Central Tendency and Dispersion Features (as described in 4.1.3), and Higher-order and Relation Statistical Features (as described in 4.1.3). Below is a brief description of the methods used for extracting these statistical features.

Central tendency and dispersion features: The extraction of Central Tendency and Dispersion Features plays a crucial role in enhancing the representation of a tabular dataset for the recognition of PD features. Features such as mean, median, variance, and standard deviation provide foundational insights into the central values and variability of the data, enabling a robust understanding of normal and anomalous patterns in the dataset[29–31]. Extremes are captured using maximum and minimum values, while the 1st and 99th percentiles, combined with the displacement between them, offer a granular view of data spread and outlier behaviour. This detailed statistical profiling helps identify subtle variations in PD-related data, such as slight tremor intensities or movement inconsistencies, which may otherwise be overlooked in raw data formats. As a result, these features enable the model to capture both average trends and deviations effectively, improving its ability to discern patterns specific to PD. Table 4 demonstrates the feature's name. The extraction of meaningful statistical features from the Kinematic features enables an insightful understanding of the dynamics within a signal. Here are the 13 statistical features calculated for each Kinematic feature. The statistical feature extraction process starts by transforming raw Kinematic features into statistically rich features that reveal patterns and trends hidden in the signal. Features like mean, median, and RMS encapsulate the central tendency and magnitude, while variance and standard deviation illuminate variability. Maximum and minimum identify extreme behaviours, and percentiles provide thresholds that uncover outliers. The statistical features and their mathematical definitions and descriptive explanations in Table 4:

Higher-order and relationship: The Higher-Order and Relationship Features capture the shape of data distributions and inter-variable relationships, enhancing the analysis. Skewness and kurtosis are key metrics: skewness indicates asymmetry in the data, while kurtosis measures the peakedness or presence of extreme values. These features are crucial for identifying abnormalities in PD patients, such as uneven gait or spastic movements, improving model sensitivity to subtle disease markers. As shown in Table 4, these features enhance

Feature name	Description
Mean	The mean represents the average value of the signal. It provides a central measure of the data, indicating the typical magnitude of the signal over its entire duration $\text{Mean}(\mu) = \frac{1}{N}\sum_{i=1}^{N} x_i$
Median	The median is the middle value of the signal when sorted in ascending order. It is a robust measure against outliers and provides a central distribution value $\text{Median} = x_{\lceil N/2 \rceil} \quad (\text{if sorted data: } x_1, x_2, \ldots, x_N)$
Variance	Variance quantifies the spread or dispersion of the signal. A higher variance indicates more variability in the signal values $\text{Variance}(\sigma^2) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$
Standard deviation	Standard deviation is the square root of variance and provides an intuitive measure of the average deviation from the mean $\text{SD}(\sigma) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$
Maximum	The maximum is the highest value within the signal. It identifies peaks or extreme values in the data. $\text{Max} = \max(x_1, x_2, \ldots, x_N)$
Minimum	The minimum is the lowest value in the signal. It captures the valleys or troughs of the data $\text{Min} = \min(x_1, x_2, \ldots, x_N)$
1st percentile	The 1st percentile is the value below which 1% of the data falls. It provides insight into the lower extremes of the distribution
99th Percentile	The 99th percentile is the value below which 99% of the data falls. It highlights the upper extremes of the distribution
Displacement between 99th and 1st percentiles	This feature measures the range between the upper and lower extremes of the data. It represents the spread of the most extreme values $Displacement = \text{Percentile}_{99} - \text{Percentile}_1$
Skewness	Skewness quantifies the asymmetry of the data distribution. Positive skew indicates a longer tail to the right, while negative skew indicates a longer tail to the left $\text{Skewness} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^3}{\sigma^3}$
Kurtosis	Kurtosis measures the "tailedness" of the data distribution. Higher kurtosis signifies more extreme outliers $\text{Kurtosis} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^4}{\sigma^4} - 3$

Table 4. Features and their descriptions.

classification accuracy by identifying complex patterns in PD datasets. Kurtosis and skewness offer deeper insights than traditional statistical measures. Kurtosis reveals the sharpness of data peaks, highlighting extreme values, while skewness indicates the direction of data asymmetry. These higher-order features, applied to handwriting movement data, allow more effective differentiation between HC and standard statistical measures. Additionally, we propose Signed Speed Features, which capture movement direction information for speed-related features, particularly those involving the direction of an arbitrary axis. After extracting a comprehensive set of statistical features from each kinematic feature, we calculated a total of 944 features, which were then fed into the subsequent stages of analysis.

Figure 5 shows the newly extracted kinematic feature with the Std statistical feature for Task-1. In the same way, Fig. 6 shows the newly extracted kinematic feature with Std statistical feature for Tasks 2, 3, 4, and Fig. 7 shows the newly extracted kinematic feature with Std statistical feature for Tasks 5, 6, 7, 8.

Feature normalization

All data used in training and validation are properly standardized prior to experimentation with machine learning models. More specifically, standardization was performed using StandardScaler and Pipeline, which were provided in the scikit-learn library. The training data mean and standard deviation were calculated, and the training and validation data were standardized using the training data mean and standard deviation.

$$\text{Normalized Value} = \frac{x-\mu}{\sigma} \tag{12}$$

where μ is mean value, σ is standard deviation.

Feature selection

In general, feature selection suppresses overfitting and increases versatility, as well as reduces computational costs and shortens training time. In the study, we selected features in two ways: Top $K\%$ features selection and SFFS. In handwriting-based tabular data analysis for dynamic movement statistical feature extraction, the selection of relevant features is critical to the performance of machine learning algorithms. High-quality features significantly enhance the accuracy and reliability of predictions, particularly in tasks related to the classification and detection of PD patients. We present boxplots for some of the features in Figs. 5, 6, and 7, highlighting newly proposed kinematic features along with the standard deviation (Std) statistical feature. These visualizations provide insights into the distribution and variability of the features, helping us assess their importance in the feature selection process. From these plots, we can observe that some features exhibit similar information, suggesting redundancy, while others show less variability and may have lower significance. This underscores the importance of careful feature selection to improve model performance by eliminating irrelevant or redundant features.

To optimize the selection process, we employed two feature selection techniques: RF and SFFS, targeting the most influential features from the handwriting data. Initially, an RF-based algorithm was utilized to rank features based on Gini impurity scores, identifying the top N% of features as the most important. The RF algorithm's ability to rank features by their importance makes it a valuable tool for reducing dimensionality while preserving the most relevant data for further analysis.

Top $K\%$ feature selection

Random forest (RF) is a supervised learning algorithm combining decision trees and ensemble learning, effective for classification and feature selection. In our study, we used an RF model to identify top-ranking features through the following steps:

- *N* subsamples were created from the original data using bootstrap sampling (random sampling with replacement), increasing model diversity.
- *N* decision trees were generated, where at each node, *M* features were randomly selected for splitting, reducing overfitting.
- The RF model was trained with combined features.
- Gini impurity was calculated for each feature, ranking them based on importance in classification.
- The top 50%, 25%, 10%, 5% and 1% of features, based on Gini impurity scores, were selected for further analysis.

This approach effectively reduced the dimensionality while retaining the most informative features for handwriting-based analysis, aiding in PD identification.

Sequential floating forward selection

In the study, we employed SFFS to select the potential features, which are shown in Fig. 8. SFFS is mainly a type of greedy algorithm to select the effective feature aiming to reduce dimensionality, which is widely used to select a subset of *k*-dimensional features from a *d*-dimensional feature space (where $k < d$) and plays a crucial role in reducing generalization errors[32]. In this study, we employed SFFS to select relevant features for PD patients across the various tasks for the following reasons: (i) This method employs the strengths of both forward and backward elimination to identify the most informative features, minimizing redundancy in the process; (ii) it initializes with an empty set, adding features incrementally to maximize model performance and interpretability, and stops once no further enhancements are achieved. The final selected feature subset is applied in ML-based gender classification (PD/HC). This approach also significantly enhances model interpretability and performance

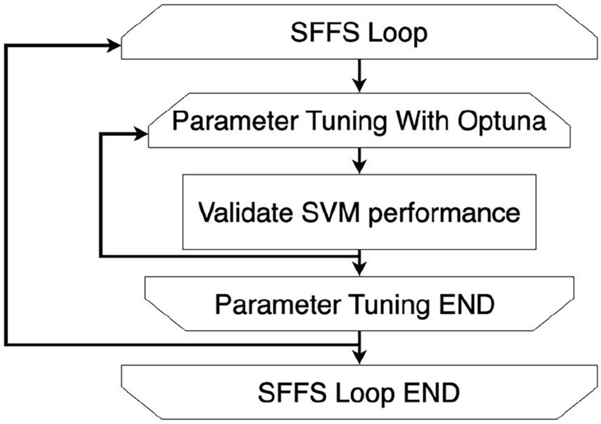

Fig. 8. SFFS architecture.

and reduces computational costs. In this experiment, we chose the Sequential Forward Selection(SFS)[32] of the Wrapper Method, and the pseudocode of the SFFS is given below:

1: **Input:** $Y = \{y_1, y_2, \ldots, y_d\}$: set of all features
2: $X_k = \{x_j \mid j = 1, 2, \ldots, k; x_j \in Y\}$, where $k \in \{0, 1, 2, \ldots, d\}$ and X_k is a subset of Y
3: **Output:** Selected feature subset X_k
4: **Initialization:** $X_0 = \emptyset$, $k = 0$
5: **repeat**
6: $\quad x^+ = \arg\max J(X_k + x)$, where $x \in Y \setminus X_k$, J is an evaluation index, and x^+ is the feature with the highest evaluation when selected.
7: $\quad X_k = X_k + x^+$
8: $\quad k = k + 1$
9: **until** k reaches the specified number of features
10: **Step 1** to **Step 3** are repeatedly iterated. When k reaches the specified number, x^+ is the set of the most appropriate features obtained. SFFS is performed up to **Step 3** of SFS, and a process for searching for features to be deleted is added. Initially, **Step 1** to **Step 4** are performed starting from $X_0 = \emptyset$, $k = 0$, as in the SFS.
11: $x^- = \arg\max J(X_k - x)$, where $x \in X_k$ and x^- is the feature with the best performance when the feature is deleted.
12: **if** $J(X_k - x) > J(X_k)$ **then**
13: $\quad X_k = X_k - x^-$
14: $\quad k = k - 1$
15: $\quad$ Go to **Step 1**
16: **end if**

Algorithm 1. SFFS algorithm.[32]

Machine learning based classifier analysis

To classify PD and HC, we developed SVM[33] classifiers trained on specific statistical features. We calculated 65 novel and 23 existing statistical features for each of the seven tasks, which include measurements pertinent to Parkinson's diagnosis. Each task-specific SVM classifier shows in Fig. 9 is optimized for performance, with parameters fine-tuned via Bayesian optimization using the Optuna library[34].

SVM classifier and kernel optimization

The SVM classifier[35,36] seeks to find an optimal hyperplane to separate PD and HC classes, given by:

$$f(\mathbf{x}) = \text{sign}\left(\sum_{\mathbf{i}=\mathbf{1}}^{\mathbf{n}} \alpha_{\mathbf{i}} \mathbf{y}_{\mathbf{i}} \mathbf{K}(\mathbf{x}_{\mathbf{i}}, \mathbf{x}) + \mathbf{b}\right) \tag{13}$$

where $\mathbf{x}_i$ are support vectors, y_i are their class labels, α_i are Lagrange multipliers, $K(\mathbf{x}_i, \mathbf{x})$ is the kernel function, and b is the bias term. We explored several kernels:

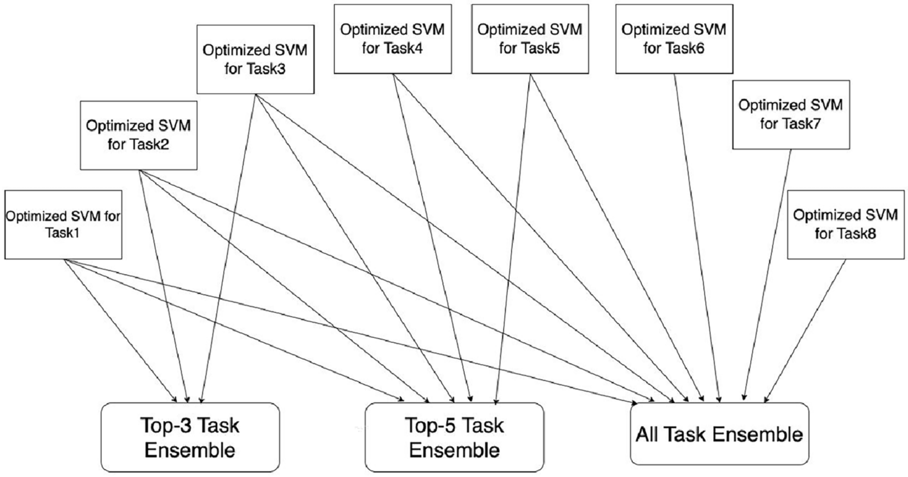

Fig. 9. Ensemble tasks classification.

- Linear kernel:

$$K(\mathbf{x}_i, \mathbf{x}) = \mathbf{x}_i \cdot \mathbf{x} \tag{14}$$

- RBF kernel:

$$K(\mathbf{x}_i, \mathbf{x}) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}\|^2\right) \tag{15}$$

- Sigmoid kernel:

$$K(\mathbf{x}_i, \mathbf{x}) = \tanh(\alpha \mathbf{x}_i \cdot \mathbf{x} + c) \tag{16}$$

The hyperparameters C and γ were optimized within the range 0.01 to 100, and the Leave-One-Out cross-validation method was used to evaluate generalization performance for each task classifier. Besides the individual task experiment, we also calculate the performance accuracy with all tasks or specific combinations of tasks, such as Top-3 or Top-5.

Individual task classification and ensemble approach
We trained an optimized SVM for each of the 7 tasks, each leveraging a different set of statistical features. After calculating the accuracy of each task-specific classifier, we proceeded to ensemble combinations of these classifiers to enhance overall classification performance.

- Top-3 task ensemble: we selected the top three task-specific SVMs with the highest individual accuracy. The ensemble output $F_3(\mathbf{x})$ is given by:

$$F_3(\mathbf{x}) = \operatorname{sign}\left(\sum_{i \in T_3} w_i f_i(\mathbf{x})\right) \tag{17}$$

where T_3 represents the indices of the top-3 performing tasks, $f_i(\mathbf{x})$ is the classifier output for the i-th task, and w_i are weights typically based on individual accuracy scores.

- Top-5 task ensemble: we extended the ensemble to include the top five task-specific SVMs. The ensemble output $F_5(\mathbf{x})$ is given by:

$$F_5(\mathbf{x}) = \operatorname{sign}\left(\sum_{i \in T_5} w_i f_i(\mathbf{x})\right) \tag{18}$$

where T_5 contains the indices of the top-5 performing tasks.

- All-task ensemble: finally, we created an ensemble using all 7 task-specific SVMs, yielding the output $F_7(\mathbf{x})$:

$$F_7(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{7} w_i f_i(\mathbf{x})\right) \tag{19}$$

Each ensemble approach was validated using Leave-One-Out cross-validation to ensure robust performance. The results showed that the ensemble methods improved classification accuracy, especially when aggregating outputs from individually optimized tasks. Figure 9 shows the ensemble task classification using SVM. Equations 13 through 19 describe the mathematical formulations of the SVM classifiers and ensemble methods used in our analysis.

Experimental results

The experimental evaluation was conducted using the PaHaW dataset, which consisted of 37 PD patients and 38 HC. Each participant completed a series of eight handwriting tasks designed to capture a range of motor skills. These tasks included: (1) drawing the Archimedes spiral; (2) writing the cursive letter "l"; (3) writing the bigram "le"; (4) writing the trigram "les"; (5) writing the cursive word "lektorka" (meaning "female teacher" in Czech); (6) writing the word "porovnat" (which means "to compare"); (7) writing "nepopadnout" (meaning "to not catch"); and (8) composing the cursive sentence "Tramvaj dnes už nepo-jede" (translating to "The tram won't go today"). To ensure the robustness of the classification results, all extracted features were normalized to achieve a zero mean and unit variance. For the purpose of evaluating model performance, we implemented Leave-One-Out Cross-Validation (LOOCV).

Leave-one-out cross-validation

LOOCV is a specific type of cross-validation technique used to assess the performance of a predictive model. In this approach, the dataset is divided into multiple subsets, where each subset consists of a single observation (data point). The model is trained on all data points except for the one left out and then tested on that single observation. This process is repeated for each observation in the dataset, resulting in as many iterations as there are observations[37–39]. The primary advantage of LOOCV lies in its ability to utilize nearly the entire dataset for training, which can lead to a more accurate estimation of the model's performance. However, it is computationally intensive, especially with larger datasets, since it requires training the model multiple times. Despite this, LOOCV can provide valuable insights into how well the model generalizes to unseen data, making it a useful technique for evaluating the performance of classifiers in studies like ours.

Experimental setup and performance metrics

This section presents the details of the experimental environment and Python/libraries. This experiment was run on a PC (with Ubuntu 22.04, Intel Core i9-13900K, 64GM RAM, Python:3.10.4, scikit-learn:1.3.0, Optuna:2.10.1, Numpy:1.23.1, Pandas:1.4.3). The performance of the trained classification models was evaluated by classification accuracy (ACC), recall, precision, and F1-Score (F1S) which are defined as follows:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \times 100[\%] \tag{20}$$

$$\text{Recall} = \frac{TP}{TP + FN} \times 100[\%] \tag{21}$$

$$\text{Precision} = \frac{TP}{TP + FP} \times 100[\%] \tag{22}$$

$$\text{F1S} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{23}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

Ablation study

In this study, we introduced novel dynamic movement statistical features and combined them with existing baseline features to form a comprehensive feature vector. To ensure the selection of the most relevant features, we applied the SFFS-based algorithm across both task-specific and ensemble tasks, leveraging state-of-the-art ML-based algorithms. For the ablation study, we further refined RF model for feature selection to rank the top k% features, and evaluated model performance using 10-fold CV. We conducted experiments by progressively selecting the top 50%, 25%, 10%, 5%, and 1% of features and comparing the accuracy across various algorithms, using both k-fold and LOOCV splitting techniques. This approach highlights the robustness of our model by revealing how feature reduction impacts performance, ensuring that the selected features offer optimal accuracy with minimal computational cost. The novelty lies in combining new dynamic features with advanced feature selection methods, demonstrating the necessity of efficient feature extraction and selection in improving classification performance.

Ablation study using all features and rf feature selection with with k-fold cross validation

Table 5 shows the ablation study, whereas we applied k-fold CV to evaluate the performance of various state-of-the-art algorithms, using the top k% selected features. The study showed how feature selection impacts model performances across different algorithms. For the SVM model, the performance improved as feature selection

Algorithms	All features	RF selected 50% features	25% of features	10% of features	5% of features	1% of features
SVM	61.43	61.25	61.79	63.04	63.04	64.29
ET	62.68	64.11	64.29	61.07	66.79	58.39
RF	64.11	63.92	65.71	62.86	60.18	68.39
GB	69.64	72.86	73.75	70.00	68.39	67.14
AB	73.57	70.89	72.32	73.57	72.32	73.57
KNN	53.04	53.04	51.79	55.71	55.89	54.11

Table 5. Performance (in %) of the ablation study with K-fold CV using all features and RF feature selection.

Algorithms	All features	Selected top50% features	Selected top 25% features	Selected top 10% features	Selected top 5% features	Selected top 1% features
SVM	63.89	62.50	65.28	66.67	65.28	63.89
ET	66.67	62.50	63.89	61.11	62.50	62.50
GB	84.72	88.89	86.11	70.83	73.61	66.67
AB	76.39	77.78	73.61	75.00	77.78	72.22
KNN	54.17	55.56	54.17	54.17	52.78	54.17

Table 6. Performance (in %) of the ablation study with LOOCV using all features and RF feature selection. *SVM* support vector machine; *ET* extra tree, *GB* gradient boost, *AB* AdaBoost.

Task number	Task name	Selected features	ACC	Precision	Recall	F1
Task 1	Spiral	Teager_Kaiser_Energy(velocity,1)	98.61	98.65	98.61	98.61
Task 2	"lll"	first_x_1st_percentile	98.66	98.72	98.65	98.67
Task 3	"le le le"	last_y_signed_velocity_min	98.66	98.68	98.68	98.67
Task 4	"les les les"	last_x_displacement_kurtosis, SNRce(displacement)	96.0	96.05	95.98	96.00
Task 5	"lektorka"	first_displacement_max, first_x_acceleration_max	98.66	98.72	98.65	98.67
Task 6	"porovnat"	first_x_signed_displacement_kurtosis	93.33	93.65	93.28	93.31
Task 7	"nepopadnout"	velocity_kurtosis	94.66	94.82	94.63	94.66
Task 8	"tramvaj dnes uz nepo-jede"	first_displacement_skewness	93.33	93.65	93.28	93.31

Table 7. Classification accuracy (in %) of the selected features for each task.

becomes more refined, reaching 64.29% with 1% of the top features. Extra Trees showed fluctuating results, peaking at 66.79% with 5% of features, demonstrating its sensitivity to the number of selected features. Random forest (RF) achieved the highest accuracy of 68.39% with 1% of features, highlighting its robustness in feature reduction. Gradient Boosting performed consistent improvements, with the best performance at 73.75% using 25% of features. AdaBoost (AB) remained stable, reaching 73.57% in both full features and 1% of the top selected features, indicating its resilience. Lastly, the KNN-based model showed modest improvements, with the highest accuracy of 55.89% using 5% of selected features.

Ablation study using all features and rf feature selection with LOOCV

In this ablation study, LOOCV was employed to evaluate the performance of various algorithms across different levels of feature selection, as illustrated in Table 6. The results highlight how selecting a percentage of the top features influences classification accuracy for different algorithms. SVM achieved the highest accuracy of 66.67% when selecting the top 10% of features, indicating an optimal balance between feature reduction and model performance. The Extra Trees algorithm displayed consistent performance across all feature levels, peaking at 66.67% accuracy when all features were included, showing robustness to feature selection. Gradient Boosting attained a classification accuracy of 88.89% when using the top 50% of features and 84.72% with all features. However, its accuracy dropped with fewer features, reaching 66.67% when only 1% of features were selected. AB also performed well, achieving 77.78% classification accuracy when selecting 50% of the features. In contrast, the KNN model exhibited modest performance compared to other algorithms, with its highest classification accuracy of 55.56% for 50% feature selection. Its performance remained stable but consistently lower than that of the other models. Overall, Gradient Boosting and AB showed improved performance with an increase in the number of selected features, while SVM and Extra Trees exhibited greater resilience to feature reduction. KNN, although less sensitive to feature selection, consistently achieved lower accuracy. These findings underscore the critical role of feature selection in enhancing model performance.

Method	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Average
Peter et al.[18]	65.4	70.0	72.3	65.4	66.7	67.7	67.1	79.4	69.25
Impedovo[22]	97.33	97.43	95.12	93.13	96.79	95.96	96.76	92.05	95.57
Proposed method	98.61	98.66	98.66	96.00	98.66	93.33	94.66	93.33	**96.99**

Table 8. Task-wise accuracy (in %) comparison of proposed system against state of the art model.

Authors	Combined tasks	Combined task name	Extracted features (total no)	FSA (selected feature no)	CA	ACC
Peter et al.[18]	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Trajectory, velocity, dynamic, acceleration, jerk, stroke (13)	None	SVM	79.4
Peter et al.[19]	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Trajectory, velocity, dynamic, acceleration, jerk, stroke , (600)	Whitney U-test filter and relief algorithm (16)	SVM	80.09
Peter et al.[20]	7	Task 2, 3, 4, 5, 6, 7, 8	Entropy, signal energy,EMD	Mann-Whitney U-test filter and relief algorithm	SVM	88.1
Peter et al.[23]	7	Task 1, 2, 3, 4, 5, 6, 7	Entropy, EMD	None	SVM	AUC-89.09
Mucha et al.[21]	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Velocity, Acceleration, Frequency domain,fractional-order derivatives	–	XGBoost	97.14
Impedovo[22]	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Baseline, 24	–	SVM	88.33
Impedovo[22]	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Baseline+new, 24+3	–	SVM	93.79
Impedovo[22]	3	Task 1, 2, 5	Baseline+new, 24+3	–	SVM	97.14
Proposed (all)	8	Task 1, 2, 3, 4, 5, 6, 7, 8	Baseline+ proposed dynamic (38)	SFFS (custom)	SVM	99.98

Table 9. State of the art accuracy (in %) comparison of combined tasks of the proposed model by combining different tasks samples. *CA* classification algorithm, *FSA* feature selection algorithm.

Performance result with selected features

We trained an SFFS-SVM model with the LOOCV protocol and optimized its parameters using Optuna while changing the feature subsets used. We identified the optimized feature subsets using classification accuracy, and their correspondence results are presented in Table 7. This table highlights the effectiveness of various features for predicting PD patients in various tasks, with most tasks demonstrating high precision, recall, and F1 scores. We observed that the best classification accuracy was obtained for one feature in most of the tasks. The classification accuracy did not improve when two or more features were considered. Each row outlined a specific task, detailing the feature name used and the performance metrics achieved. Notably, Task 2, Task 3 and Task 5 achieved the highest accuracy of 98.66% . Whereas, Task 6 recorded the lowest accuracy of 93.33%.

State of the art comparison for PD vs non-PD classification performance

We performed a comparison study of our proposed method against previous models published in two ways: task-wise and combined task with sample-wise, which are briefly explained in the following “Task-wise state of the art comparison” and “Combined task-wise state of the art comparison”.

Task-wise state of the art comparison

Table 8 highlights the efficacy of our proposed method in comparison to existing approaches, particularly in maintaining high classification accuracy across multiple tasks. As indicated in Table 8, Peter et al.[18] proposed a PD detection model and achieved an average classification accuracy of 69.25%, with the highest accuracy of 79.4% for Task 8, whereas the lowest classification accuracy of 65.4% for Task 1[18]. In contrast, Donato Impedovo[22] demonstrated superior performance, achieving an average accuracy of 95.57% and the best detection rate of 97.33% for Task 1, whereas Task 4 provided the lowest accuracy of 93.13%[22]. Our proposed method outperformed compared to existing models[18,22]. More specially, our proposed method obtained an average classification accuracy of 96.99%, with the highest accuracy of 98.66% in both Task 2 and Task 3, while the lowest classification of 93.33% was obtained for both Task 6 and Task 8.

Combined task-wise state of the art comparison

Table 9 compares the state-of-the-art performance of various models, including the proposed model, across combined tasks, focusing on accuracy and methodologies. Each column represents critical aspects of the approaches, such as the method used, the number of combined tasks, the names of those tasks, the feature types extracted (along with their total number), the feature selection algorithm utilized (with the count of selected features), the classification algorithm applied, and the accuracy achieved. This result outperforms the current state-of-the-art model. Peter et al. explored different task combinations and achieved classification accuracies ranged from 79.4% to 88.1%, obtained by SVM and various feature selection methods[18–20]. Mucha et al. also utilized advanced featuree and obtained a classification accuracy of 97.14% using XGBoost[21]. Impedovo et al. showed performance using both baseline and newly proposed features and obtained the classification accuracies between 88.33% and 97.14%[22]. In contrast, the proposed method demonstrates superior performance, achieving an impressive accuracy of 99.98% across various task combinations. This is accomplished by employing a

Diagnosis-age	Number of patient	Pateint ID no.				Diagnosis-age	Number of patient	Pateint ID no.		
1	1	77	–	–	–	9	2	6	23	74
2	4	5	13	53	78	10	3	25	54	–
3	3	9	10	36	–	12	2	4	48	75
4	3	16	33	43	–	14	2	22	55	–
5	4	8	34	44	80	17	1	15	18	–
6	3	1	19	98	–	18	1	17	–	
7	2	3	24	–	–	19	0	7	–	
8	3	2	14	20	–	–	–	–	–	

Table 10. Diagnosis age and corresponding patient IDs[40].

Task No	Selected feature list	Accuracy	Parameters	Best features
Task-1	pressure_standard_deviation’, ’stroke_pressure_1st_each_stroke 99th_percentile’, ’x_velocity_no_abs_median’	88.88	C’: 434.78, ’kernel’: ’sigmoid’, ’gamma’: 855.93, ’degree’: 3, ’coef0’: 309.43, ’random_state’: 42	X,Y, pressure, velocity, displacement, kurtosis, skewness
Task-2	stroke_pressure_std each_stroke_standard_deviation	83.78	C’: 0.56, ’kernel’: ’sigmoid’, ’gamma’: 486.44, ’degree’: 3, ’coef0’: 695.74, ’random_state’: 42	Stroke, azimuth, altitude, Y, displacement, X
Task-3	velocity_max	81.08	C’: 0.94, ’kernel’: ’sigmoid’, ’gamma’: 828.14, ’degree’: 2, ’coef0’: 995.42, ’random_state’: 42	Velocity, X,Y,displacement, azimuth,stroke
Task-4	stroke_pressure_std each_stroke_standard_deviation	86.48	’C’: 956.75, ’kernel’: ’sigmoid’, ’gamma’: 232.37, ’degree’: 2, ’coef0’: 53.91, ’random_state’: 42	Stroke, X, displacement, velocity, Y, pressure
Task-5	last_altitude_99th_percentile’, ’last_x_velocity_min’	89.18	C’: 0.51, ’kernel’: ’sigmoid’, ’gamma’: 780.26, ’degree’: 1, ’coef0’: 849.56, ’random_state’: 42	Altitude, X, veloicty, altitude, stroke, entropy
Task-6	Renyi_Entropy(y,2)’, ’teager_kaiser_energy(displacement,1)’	86.48	C’: 801.73, ’kernel’: ’sigmoid’, ’gamma’: 639.29, ’degree’: 4, ’coef0’: 242.12, ’random_state’: 42	Entropy, displacement, stroke, Y, velocity, X
Task-7	x_velocity_skewness’, ’x_velocity_no_abs_skewness’, ’first_y_max’	89.18	C’: 291.21, ’kernel’: ’sigmoid’, ’gamma’: 627.59, ’degree’: 5, ’coef0’: 711.93, ’random_state’: 42	X, Y, velocity, pressure, displacement, stroke
Task-8	y_displacement_between 99th_percentile_and_1st_percentile’	86.48	C’: 846.66, ’kernel’: ’sigmoid’, ’gamma’: 550.41, ’degree’: 1, ’coef0’: 912.67, ’random_state’: 42	Y, displacement, X, pressure, velocity, angle

Table 11. Performance evaluation of Parkinson’s disease classification across tasks using SVM and SFFS for early, mid, and late stage PD patients.

combination of baseline and custom dynamic features. The important features were selected by the SFFS-based algorithm. The use of SVM for classification further enhances the robustness of the proposed methodology, highlighting its strength in effectively distinguishing complex patterns within handwriting data for PD detection.

Classification of PD patients in early, mid, and late stages based on the age of the disease

To further investigate the potential of the proposed feature extraction and classification methodology, we classified the 37 Parkinson’s Disease (PD) patients shows in the Table 10 and Fig. 1 into three distinct stages: early stage, mid stage, and late stage. This classification was based on the patients’ years since diagnosis, allowing for a clear age-based distribution of the stages. The stages were defined as follows: Early Stage (1 to 4 years of diagnosis) included 11 patients, distributed as 1, 4, 3, 3 across different time periods. The Mid Stage (5 to 8 years of diagnosis) consisted of 12 patients, distributed as 4, 3, 2, 3 across different time periods. The Late Stage (9 to 19 years of diagnosis) included 14 patients. A total of 14 patients were classified into the Late Stage (diagnosed 10+ years ago), providing a sufficient representation of PD patients with prolonged disease progression. Table 11 shows the Performance Evaluation of Parkinson’s Disease Classification Across Tasks Using SVM and Sequential Forward Floating Selection (SFFS) for Early, Mid, and Late-stage PD Patients. The table includes the selected features, classification accuracy, hyperparameters, and the most important features for each task.

Feature extraction and selection for stage classification

The feature extraction process was identical to the one used for general PD classification. Dynamic kinematic features and hierarchical statistical features were extracted from the motor data. To identify the most relevant features for distinguishing between the early, mid, and late stages of PD, the Sequential Forward Floating Selection (SFFS) method was applied. This approach reduced dimensionality and computational complexity while preserving the critical features that contribute to stage differentiation.

Average of three-stage based classification performance

The performance of the Parkinson’s disease (PD) classification model across early, mid, and late stages was evaluated by calculating the average accuracy for each task. For most tasks, the model demonstrated strong overall classification accuracy, with Task-5 achieving the highest average accuracy of 89.18%. This task, utilizing features such as altitude and velocity, showed particularly effective performance in distinguishing PD stages, especially in the later stages of the disease. Task-1 and Task-7, which focused on pressure and velocity skewness, also recorded high average accuracy (88.88% and 89.18%, respectively), indicating the key role of these kinematic features in capturing motor impairments across all stages of PD. However, tasks like Task-2 and Task-3, which primarily focused on features like stroke pressure standard deviation and velocity, showed slightly lower average accuracy (83.78% and 81.08%), suggesting that these features may be less effective for distinguishing between stages in more subtle or complex cases. In general, tasks involving displacement, pressure, and velocity features consistently achieved higher accuracy, underscoring their importance in differentiating PD stages. Despite the strong performance, tasks like Task-3 (velocity) and Task-2 (pressure standard deviation) displayed somewhat lower accuracy, particularly in the mid-stage, where motor symptoms overlap more between stages. The results highlight the challenges in accurately classifying mid-stage PD, where subtle variations may be more difficult to capture. Overall, the average accuracy of the model across all tasks indicates the effectiveness of the proposed methodology in distinguishing early, mid, and late stages of Parkinson's Disease, with specific tasks excelling in identifying motor impairments across the stages.

Discussion

This study presented an optimized methodology for detecting PD that effectively captures movement dynamics in handwriting tasks. We extracted features from each kinematic feature in two ways: kinematic and statistical-based features. We extracted kinematic-based features, including angle trajectory, signed x/y displacements, velocity, and first/last displacements. Unlike conventional methods that analyze the handwriting task, our methodology focuses on dynamic changes within the first and last 10% of the task. Moreover, we also extracted some statistical features such as kurtosis, skewness, and angular measures. This enhanced the analysis by capturing subtle variations in speed and directional movement. The findings showed that our proposed system achieved a classification accuracy of 96.99% for task-wise evaluation, as shown in Table 8, and 99.98% classification accuracy for ensemble tasks, as shown in Table 9. This performance accuracy is around 2% improvement over existing state-of-the-art models. First, new features were proposed, including the dynamic movement of the hand with the existing baseline features that enhance the feature dimension and create unique patterns. Second, introducing kurtosis and skewness indices proved valuable. These features reflect differences in handwriting speed variability, a known characteristic of PD patients. While healthy subjects maintain relatively stable writing speeds, patients with PD exhibit greater variability, resulting in measurable deviations in distribution and peak sharpness. Lastly, we employed ensemble learning with SVMs trained on single features to mitigate feature interference, a challenge observed when training with multiple feature vectors simultaneously. In the SVM model, the optimization of kernels played a crucial role in enhancing the classification accuracy. In our study, we observed that SVM with a sigmoid kernel performed better compared to other kernels. Previous studies have commonly used RBF or linear kernels. By using individual SVMs for each feature and combining their outputs through ensemble learning, we ensured that features did not interfere with one another. Additionally, our method classifies PD patients into distinct stages—Early, Mid, and Late—based on the age of the disease, enhancing the understanding of PD's evolution over time. This contrasts with typical PD detection methods, which only differentiate between PD and non-PD. Our approach leverages the strengths of each SVM while minimizing the potential for conflicts among features, enhancing the overall performance and robustness of the model. Finally, our methodology outperforms existing models and provides a robust, reliable framework for PD detection. The combination of innovative feature extraction, advanced SVM optimization, and ensemble learning sets a new benchmark in handwriting-based diagnostic tools, offering promising applications for clinical use globally.

Implications for medical practice

Our proposed system offers significant potential for aiding medical professionals in the accurate detection of PD, both in Japan and globally. In Japan, where the ageing population is driving the prevalence of neurodegenerative diseases like PD, our system provides a non-invasive, cost-effective, and objective diagnostic tool. By analyzing handwriting data, doctors can detect early signs of PD that may be missed during standard clinical examinations. The high classification accuracy reduces the risk of misdiagnosis, enabling early intervention, which improves patient outcomes and reduces the societal burden of the disease. The system's adaptability to different languages and cultural contexts makes it suitable for deployment in other countries, enhancing global access to PD detection tools. In regions with limited access to advanced neuroimaging facilities or specialized neurological care, this system can serve as an affordable and reliable alternative. By equipping healthcare professionals with actionable insights derived from handwriting analysis, our approach has the potential to revolutionize PD detection and contribute to more personalized and effective patient care worldwide.

Real-time performance

Real-time application in clinical settings is a significant challenge. Although the system demonstrates high accuracy in controlled environments, clinical practice involves several variables, such as varying handwriting speeds, tremors, and other motor impairments that could affect handwriting performance. We aim to optimize the system for real-time analysis with minimal latency, ensuring quick and actionable insights for healthcare providers.

Range of PD patients and patient variability

The method has demonstrated excellent performance for early to moderate stages of PD, but its applicability to more advanced stages (Stages 4 and 5) remains to be explored. As motor impairments worsen, handwriting may become significantly more impaired, potentially affecting the system's ability to detect PD symptoms. Future work will investigate how the method can be adapted for more severely affected patients. Variability in motor skills, cognitive impairments, and medication states may also impact the system's performance. For example, PD patients on dopaminergic medication (L-DOPA) may experience fluctuations in handwriting quality due to medication effects. Therefore, the system must be robust enough to accommodate variations in motor performance over time and across different patients. Additionally, different handwriting tasks may reveal different aspects of motor impairment. Tasks such as drawing the Archimedean spiral or writing specific letters and words may correlate differently with PD symptoms like bradykinesia, rigidity, and tremors. Future work will involve a more detailed analysis of task-specific performance to optimize task selection for various clinical settings.

Global applicability

The system's adaptability to different languages and cultural contexts is crucial for global deployment. All participants in this study were native Czech speakers, and future studies will need to test the system in diverse linguistic and cultural settings to ensure its effectiveness across populations. This will help ensure that the system is suitable for broad use across global populations, thereby enhancing its applicability and utility in PD detection.

Conclusion

This study proposed an optimized methodology for PD detection that integrated newly developed dynamic kinematic-based features with advanced ML-based techniques to capture the movement dynamics during handwriting tasks. Extracting 65 novel kinematic features which allow us to capture subtle movements that traditional methods may overlook. Furthermore, we reused 23 existing kinematic features, resulting in a comprehensive set of features designed to improve the accuracy of PD detection. To further refine our approach, we enhance kinematic features using statistical formulas to compute hierarchical-based features, allowing us to better capture subtle movement variations that differentiate patients with PD from HC. We then optimised the feature set by applying the SFFS-based method to reduce the dimensionality and computational complexity. Our proposed ML-based ensemble method obtained outstanding results, with a classification of 96.99% for task-wise evaluations and 99.98% for task ensemble. This represents a 2% improvement in classification accuracy compared to the previous models. Additionally, our method classifies PD patients into distinct stages—early, mid, and late—based on the age of the disease, enhancing the understanding of PD's evolution over time. This contrasts with typical PD detection methods, which only differentiate between PD and non-PD. These outstanding results highlight the potential of our proposed approach to redefine benchmarks for PD detection, offering significant improvements in both classification accuracy and robustness. In future work, we aim to expand our model by incorporating additional datasets, including both PD patients and healthy controls, to evaluate its generalizability across different populations. Furthermore, we plan to explore real-time deployment of our proposed system, which could provide valuable insights into its practical application for PD diagnosis in clinical settings.

Data availability

PaHaW dataset source: https://www.sciencedirect.com/science/article/abs/pii/S0933365716000063?via%3Dihub Download Procedure: https://bdalab.utko.fekt.vut.cz/ Code and Dataset: https://github.com/musaru/PD_PaHaW

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Author contributions

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Declarations

Competing interests

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Additional information

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